Counting all decimal Kaprekar three-cycles

A computer-assisted proof of the eighteen-branch formula

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October 31, 2026

Abstract

We classify the exact period-three orbits of the ordinary decimal Kaprekar map, without preserving leading zeros between iterations. Every such orbit belongs to one of the two families displayed by Kauers and Koutschan in 2023. We first classify the cycles whose extreme mirrored differences are always 9 and 1, and use zero-count monotonicity to reduce all remaining possibilities to a finite formula in unbounded integer linear arithmetic. An SMT refutation excludes those possibilities. A separate checker validates its proof graph through scoped logical reasoning and strictly checked local arithmetic proofs from a second solver. The family counts then prove all eighteen polynomials in their Conjecture 23, with zero cycles at lengths one and two. We give the arithmetic encoding and its proof, the orbit correspondence, the exact generating-function identity, and reproducibility material. The result is computer-assisted and is not a Lean formalization.

1 Statement and prior work

For a nonnegative integer x, write its ordinary decimal expansion, sort its digits into decreasing and increasing order, and subtract the second integer from the first. Denote the result by $T(x)$. Initial zeros in the increasing string are allowed, but the next iteration uses the ordinary expansion of $T(x)$, without artificial padding. Let $a(n)$ count orbits of *exact* period three consisting of n-digit integers. A cyclic rotation of an orbit is not counted again. Length cannot increase under T, so every periodic orbit has constant length.

Kauers and Koutschan [1, Section 6.8, Conjecture 23] conjectured eighteen degree-five formulas for $a(n)$, the OEIS sequence A164735 [2]. They also exhibited two families, called X and Y, whose counts give those formulas. What remained unproved was exhaustiveness: an additional family could invalidate the formulas at a larger length. The families and the formulas are thus prior work. The contribution here is the exhaustive exclusion, together with its explicit arithmetic model and reproducible verification.

Kay and Downes-Ward [3, Theorem 2, Theorems 5 and 9, Example 10] study digit-count structure and symmetric cycles in even bases. Their exhaustive classification is for base four, and does not supply the all-length decimal exclusion needed here. A bounded literature audit on October 31, 2026, including the current OEIS entry and the public `openai/math` contents, located no prior complete proof of this specific conjecture. This is not a claim of an exhaustive worldwide novelty search.

Theorem 1. *Every ordinary decimal Kaprekar orbit of exact period three belongs to one of the families X and Y defined in Section 3. Moreover, $a(1) = a(2) = 0$, and for $n = 18k + i \geq 3$, $k \in \mathbb{Z}_{\geq 0}$, $0 \leq i \leq 17$,*

$$a(n) = P_i(k)/40,$$

where the eighteen polynomials P_i appear in Appendix A. Equivalently,

$$\sum_{n\geq 1} a(n)q^n = \frac{q^8}{3(1-q^9)}\left(\frac{1}{(1-q^2)^5} - \frac{1}{(1-q^2)^2(1-q^6)}\right) + \frac{q^8}{(1-q^2)^3}. \tag{1}$$

The exhaustive-classification part of the proof is computer-assisted.

2 Sorting, subtraction and histogram dynamics

Write the sorted digits of an n-digit input as $D_1 \geq \cdots \geq D_n$. Set $r = \lfloor n/2 \rfloor$ and $d_i = D_i - D_{n+1-i}$ for $1 \leq i \leq r$. These differences are nonincreasing. If they are all zero, every digit of the input is equal, and $T(x) = 0$, which cannot belong to an exact three-cycle. Otherwise let s be the number of positive differences, let $p = d_1$, $b = d_s$, and $t = n - 2s$.

Lemma 2 (Subtraction word)**.** *At width n, the digits of the subtraction result are*

$$d_1 \cdots d_{s-1}(d_s - 1)\, 9^t\, (9 - d_s) \cdots (9 - d_2)(10 - d_1). \tag{2}$$

Here the first and reversed middle strings are empty when $s = 1$. The result has fewer than n digits if and only if $s = 1$ and $p = 1$.

Proof. The raw column differences from the most significant to the least significant end are $d_1, \ldots, d_s$, then zeros, then $-d_s, \ldots, -d_1$. The rightmost negative column borrows once, giving $10 - d_1$. Every subsequent negative column already owes one, giving $9 - d_i$. The borrow passes through all central zero columns, turning them into nines, and is discharged at d_s, changing it to $d_s - 1$. All earlier positive columns remain unchanged. If $s \geq 2$, the first digit is $d_1 \geq 1$. If $s = 1$, the first digit is $p - 1$, which vanishes exactly when $p = 1$. □

Let $c = (c_0, \ldots, c_9)$ be a digit histogram. Define

$$L_u = \sum_{j<u} c_j, \qquad R_v = \sum_{j>v} c_j,$$

and for $0 \leq u < v \leq 9$ put

$$M_{u,v}(c) = \max\{0, \min(L_u + c_u, R_v + c_v) - \max(L_u, R_v)\}. \tag{3}$$

For $1 \leq j \leq 9$, set

$$h_j = \sum_{u=0}^{9-j} M_{u,u+j}(c), \qquad h_0 = 0, \quad s = \sum_{j=1}^{9} h_j, \quad t = \sum_{j=0}^{9} c_j - 2s. \tag{4}$$

The convention $h_0 = 0$ is deliberate: all zero differences and a possible central unpaired digit are represented by t.

Lemma 3 (Exact histogram transition)**.** *The integer h_j in (4) counts positive mirrored differences equal to j. Let $p = \max\{j : h_j > 0\}$ and $b = \min\{j : h_j > 0\}$. For a full-width subtraction the output histogram $F(c)$ is given, for $0 \leq j \leq 9$, by*

$$F(c)_j = h_j + h_{9-j} + t\mathbf{1}_{\{j=9\}} - \mathbf{1}_{\{j=b\}} + \mathbf{1}_{\{j=b-1\}} - \mathbf{1}_{\{j=9-p\}} + \mathbf{1}_{\{j=10-p\}}. \tag{5}$$

Full width is equivalent to $h_1 = 0$ or $s \geq 2$.

Proof. The occurrences of digit u in ascending order occupy the half-open interval $[L_u, L_u + c_u)$, whereas digit v in descending order occupies $[R_v, R_v + c_v)$. The overlap length is (3). When $u < v$ every overlap position lies in the first half: beyond the midpoint the ascending digit is at least the descending digit. Thus these overlaps count each positive mirrored pair once. The unpaired central block has length t. In (2), the base pairs contribute digits d_i and $9 - d_i$. The two endpoint changes replace b by $b - 1$ and $9 - p$ by $10 - p$, giving (5). The last assertion is Lemma 2, since the existence of p, b already requires $s > 0$. □

Lemma 4 (Orbit correspondence)**.** *Full-width exact three-cycles of F correspond bijectively, up to cyclic rotation, to exact three-cycles of T of the same length.*

Proof. Let $K(c)$ be the integer obtained by sorting and subtracting a histogram. If $c^{(0)}, c^{(1)}, c^{(2)}$ is a histogram cycle, put $x_i = K(c^{(i-1)})$, with indices modulo three. Then $\mathrm{hist}(x_i) = c^{(i)}$ and $T(x_i) = x_{i+1}$. Distinct histograms give distinct integers. Conversely, two members of an integer cycle cannot have the same histogram: they would have the same successor, whereas T is injective on a cycle. These constructions are inverse, including their rotations. A period dividing three is either one or three, so excluding equal consecutive histograms excludes exactly the fixed points. □

3 The two families and their counts

We use difference multiplicities to describe the published families. Let $\epsilon = n \bmod 2$, and let $D = (D_0, D_1, \ldots, D_9)$ count all mirrored differences, including zero. Thus $n = 2\sum D_j + \epsilon$. The map on these vectors consists of Lemma 2 followed by pairing the sorted output digits. The output word is determined by D and ϵ. The proof of Lemma 4 also applies to this representation on cycles.

3.1 Family X

Choose $s, y \geq 0$, $x, z, w, e \geq 1$, and $\epsilon \in \{0, 1\}$. Put $m = 2s + \epsilon$ and

$$D = (s, x, m, y, m, z, m, w, m, e). \tag{6}$$

In the original word notation of [1], the corresponding parameters are $(m, a, b, c, d, e) = (m, x - 1, y, z, w, e)$. The maximum and minimum positive differences are 9 and 1, so the endpoint corrections cancel. The output histogram is

$$(e, x + m, w + m, y + m, z + m, z + m, y + m, w + m, x + m, e + m). \tag{7}$$

The successive unequal pairs in that histogram are

pair	$(9,0)$	$(9,1)$	$(8,1)$	$(8,2)$	$(7,2)$	$(7,3)$	$(6,3)$	$(6,4)$	$(5,4)$
count	e	m	x	m	w	m	y	m	z

followed by s pairs $(5, 5)$ and, when $\epsilon = 1$, one central 5. Consequently the transition replaces (x, z, w) by (z, w, x). The period is one precisely when $x = z = w$, and otherwise it is three. Its length is

$$n = 18s + 9\epsilon + 2(x + y + z + w + e). \tag{8}$$

There is no duplication of parameters in a difference vector. Every nonfixed orbit contains three such vectors. Hence the generating function for X orbits is

$$X(q) = \frac{q^8}{3(1 - q^9)} \left(\frac{1}{(1 - q^2)^5} - \frac{1}{(1 - q^2)^2 (1 - q^6)} \right). \tag{9}$$

Indeed, four variables are positive and one is nonnegative. The fixed-point subtraction forces three of the positive variables equal; and $\sum_{s \geq 0, \epsilon \in \{0,1\}} q^{18s+9\epsilon} = 1/(1 - q^9)$.

For the classification test it is useful to recognize this family directly from a histogram. Define $\mathcal{X}(c)$ to mean, with $m = c_9 - c_0$,

$$\begin{gathered} m \geq 0, \quad c_0 \geq 1, \quad c_j = c_{9-j} \ (1 \leq j \leq 4), \\ c_1, c_2, c_4 \geq m + 1, \qquad c_3 \geq m. \end{gathered} \tag{10}$$

These conditions are equivalent to (7), and recover its parameters uniquely. Taking $\epsilon = m \bmod 2$ and $s = (m - \epsilon)/2$ recovers a vector of form (6) whose output histogram is c. Thus $\mathcal{X}$ is invariant under the transition; its nonfixed states are exactly the X three-cycles.

3.2 Family Y

Choose $b, u \geq 1$, $v \geq 0$. The three difference vectors are

$$\begin{aligned} A &= (0,0,0,b,0,u,1,v,1,0), \\ B &= (0,u,0,b,1,v,1,0,0,0), \\ C &= (0,v+1,1,b,0,0,0,u-1,1,0), \end{aligned}$$

with even length $n = 2(b+u+v+2)$. Formula (5) gives their respective output histograms

$$\begin{aligned} H_A &= (0,0,v+2,b,u,u,b+1,v,1,0), \\ H_B &= (1,u-1,0,b,v+2,v+1,b+1,0,u,0), \\ H_C &= (1,v,u+1,b,0,0,b,u,v+2,0). \end{aligned}$$

Pairing H_A gives $(8,2)$ once, $(7,2)$ v times, $(6,2)$ once, $(6,3)$ b times and $(5,4)$ u times, so the next vector is B. Pairing H_B gives $(8,0)$ once, $(8,1)$ $u-1$ times, $(6,3)$ b times, $(6,4)$ once and $(5,4)$ $v+1$ times, so the next is C. Pairing H_C gives $(8,0)$ once, $(8,1)$ v times, $(8,2)$ once, $(7,2)$ u times and $(6,3)$ b times, so the next is A. Their positive extreme differences are respectively $(3,8),(1,6),(1,8)$; the states are distinct and every orbit has exactly one A. Parameters are recovered from A, proving uniqueness. These orbits are disjoint from X, whose difference-9 multiplicity is positive. Their generating function is

$$Y(q) = \frac{q^8}{(1-q^2)^3}. \tag{11}$$

Define the predicate $\mathcal{Y}(c)$ by

$$\begin{aligned} c_0 = 1, \quad c_2 = c_7 = c_9 = 0, \quad c_8 = c_1 + 1, \\ c_6 = c_3 + 1, \quad c_4 = c_5 + 1, \quad c_3 \geq 1, \quad c_5 \geq 1. \end{aligned} \tag{12}$$

This specifies exactly H_B, with $u = c_1 + 1$, $b = c_3$, $v = c_5 - 1$. The corresponding original $Y_{a,b,c}$ parameters are $(u-1, b, v)$. Each Y orbit contains exactly one such histogram: H_A has $c_0 = 0$ and H_C has $c_2 = u + 1 > 0$.

4 Exhaustive classification by integer arithmetic

The following finite arithmetic formula searches for a counterexample to the classification. All its variables are *unbounded integers*. There is no maximum length or maximum digit multiplicity.

Introduce three ten-component vectors $c^{(0)}, c^{(1)}, c^{(2)}$ with nonnegative coordinates, each of the same sum $n \geq 2$. At each phase $i \in \{0,1,2\}$ introduce $h_1^{(i)}, \ldots, h_9^{(i)}, t_i, p_i, b_i$, and impose (3)–(4), followed by

$$t_i \geq 0, \quad 1 \leq b_i \leq p_i \leq 9, \tag{13}$$

$$(j < b_i \text{ or } j > p_i) \implies h_j^{(i)} = 0, \qquad 1 \leq j \leq 9, \tag{14}$$

$$(j = b_i \text{ or } j = p_i) \implies h_j^{(i)} \geq 1, \qquad 1 \leq j \leq 9, \tag{15}$$

$$h_1^{(i)} = 0 \text{ or } \sum_{j=1}^{9} h_j^{(i)} \geq 2. \tag{16}$$

Require (5) with output $c^{(i+1)}$, indices modulo three, and finally require

$$c^{(0)} \neq c^{(1)}, \qquad \neg\mathcal{X}(c^{(0)}), \qquad \bigwedge_{i=0}^{2} \neg\mathcal{Y}(c^{(i)}). \tag{17}$$

Call the conjunction of all these constraints Φ.

Lemma 5. *Φ is satisfiable if and only if there exists an ordinary decimal Kaprekar three-cycle outside families X and Y.*

Proof. For a cycle outside the families, take its histograms and true mirrored difference multiplicities. Lemma 3 supplies every edge constraint. Conditions (13)–(15) state exactly the support endpoints, while (16) enforces the ordinary, non-padded convention. An orbit outside X has no X member by invariance, and an orbit outside Y has no $\mathcal{Y}$ member. Thus (17) holds.

Conversely, a satisfying assignment gives actual histograms with positive mirrored differences. In particular, the all-zero digit histogram and every constant-digit histogram are excluded. A positive integer with each prescribed histogram exists by placing a nonzero digit first. Lemma 3 gives a full-width histogram cycle, and Lemma 4 lifts it uniquely to an exact three-cycle of integers. The negated family predicates exclude exactly the two known possibilities. Thus a satisfying assignment is a genuine counterexample. □

Lemma 6 (Computer-assisted exclusion)**.** *The formulas Φ and Ψ are unsatisfiable over the integers, where Ψ is the reduced formula in (25).*

Verification procedure. Appendix B proves that a cycle having at least two zero digits is regular at every phase, and that every such regular cycle belongs to X. Its coverage lemma shows that a counterexample outside X and Y, if one exists, has all zero counts at most one and can be rotated to have a nonregular first phase. The same appendix proves the uniform inner-digit imbalance bound. Consequently adding exactly (25) preserves satisfiability: Φ has a solution if and only if Ψ does.

Replace $\min(a, b)$ by the conditional expression `if` $a \le b$ `then` a `else` b, and similarly for max. Replace each indicator by a conditional 1 or 0. All remaining expressions are linear integer arithmetic with Boolean combinations; there is no variable multiplication, floating-point approximation or digit-length bound. The supplement constructs Ψ directly and exports the exact formula in SMT-LIB format. Z3 [4], with proof production and arithmetic engine 2, returns `unsat` and a proof graph ending in false.

A separate verifier reads the original exported assertions and checks every reachable node of that graph. It tracks the open hypotheses of each node, validates elementary propositional and equality inferences directly, and checks every remaining local implication with cvc5 [5] under logic `QF_LIA`. These local cvc5 checks require complete internal proof checking and reject proof nodes marked `TRUST`, `HOLE` or `UNKNOWN`. The root must conclude false with no undischarged hypotheses. Appendix C describes the scoped verification rule and identifies the exact query and proof files.

Acceptance of the resulting certificate proves Ψ unsatisfiable. The analytic coverage lemma then proves Φ unsatisfiable as well. This is an unbounded integer-arithmetic exclusion; the finite example checks are not used in place of it. The trusted computational components include the query construction, serialization and parsing, the separate proof graph verifier, and cvc5's internal proof-checking implementation. Z3 supplies the candidate refutation, whose individual inferences are checked by that separate procedure. This is not a Lean formalization or a claim of external human peer review. □

5 Counting and the eighteen formulas

Lemmas 5 and 6 establish exhaustiveness. The disjoint family counts (9) and (11) now prove (1). Their sum also equals the rational generating function conjectured by Chai Wah Wu in the OEIS entry [2]. For completeness we explain the exact algebraic identification with the eighteen formulas, without appealing to a fit. Write $P_i(k) = \sum_{j=0}^{5} p_{ij} k^j$ and define

$$S_0(z) = \frac{1}{1-z}, \qquad S_{j+1}(z) = z \frac{d}{dz} S_j(z).$$

As formal power series, $S_j(z) = \sum_{k\geq 0} k^j z^k$, with the $j = 0$ case including $k = 0$. Set

$$B(q) = \frac{1}{40} \sum_{i=0}^{17} q^i \sum_{j=0}^{5} p_{ij} S_j(q^{18}) - 3 - q^2. \tag{18}$$

The two subtracted terms remove the polynomial extensions at $n = 0$ and $n = 2$, which lie outside the conjectured domain. The extension at $n = 1$ is already zero. Clearing denominators gives the exact identity

$$B(q) = X(q) + Y(q).$$

This can be verified using only polynomial arithmetic over $\mathbb{Q}$: each $(1 - z)^6 S_j(z)$ is a polynomial, and substitution of the table in Appendix A makes both cross-multiplied numerators identical. The supplied verifier performs this comparison coefficient by coefficient. It proves all eighteen identities for all k in their stated domains.

Finally, a one-digit input maps to zero. For a two-digit input with unequal digits, let their difference be $d \in \{1, \ldots, 9\}$. Its image is $9d$. At $d = 1$ this has one digit. For $2 \leq d \leq 9$, the next difference is $|2d - 11|$, whose values in order are $7, 5, 3, 1, 1, 3, 5, 7$. All reach $d = 1$, so no two-digit periodic orbit of length three exists. This completes Theorem 1.

The first nonzero example is

$$64308654 \longmapsto 83208762 \longmapsto 86526432 \longmapsto 64308654.$$

Some resulting counts are

n	8	10	12	14	16	18	19	20	21	22
$a(n)$	1	4	10	20	36	60	1	94	4	141.

These values illustrate the theorem; they are not used to prove exhaustiveness.

A The eighteen polynomials

These are the numerators from Kauers–Koutschan Conjecture 23, retained in their factored form. In each row $a(18k + i) = P_i(k)/40$, provided $18k + i \geq 3$.

i	$P_i(k)$
0	$3(243k^5 + 405k^4 + 35k^3 + 395k^2 - 318k + 40)$
1	$k(729k^4 - 405k^3 - 615k^2 + 225k + 106)$
2	$729k^5 + 1620k^4 + 735k^3 + 1320k^2 - 684k + 40$
3	$k(729k^4 - 705k^2 + 136)$
4	$3k(243k^4 + 675k^3 + 515k^2 + 565k - 118)$
5	$k(729k^4 + 405k^3 - 615k^2 - 225k + 106)$
6	$3k(243k^4 + 810k^3 + 845k^2 + 790k + 32)$
7	$3k(k + 1)(243k^3 + 27k^2 - 142k + 12)$
8	$729k^5 + 2835k^4 + 3705k^3 + 3405k^2 + 726k + 40$
9	$3k(k + 1)(243k^3 + 162k^2 - 127k - 18)$
10	$729k^5 + 3240k^4 + 5055k^3 + 4860k^2 + 1636k + 160$
11	$3k(k + 1)(243k^3 + 297k^2 - 52k - 48)$
12	$729k^5 + 3645k^4 + 6585k^3 + 6795k^2 + 2926k + 400$

i	$P_i(k)$ (continued)
13	$3k(k+1)(243k^3+432k^2+83k-58)$
14	$729k^5+4050k^4+8295k^3+9270k^2+4696k+800$
15	$3k(k+1)(243k^3+567k^2+278k-28)$
16	$3(k+3)(243k^4+756k^3+1127k^2+734k+160)$
17	$3k(k+1)(243k^3+702k^2+533k+62)$

B Analytic classification of regular phases and a reduced query

Call a full-width histogram transition *regular* when its maximum positive mirrored difference is $p = 9$ and its minimum is $b = 1$. The following argument proves that any cycle regular at every phase belongs to X. It also explains exactly which additional constraints may be imposed on the remaining integer-arithmetic query.

Lemma 7 (Zero-count reduction)**.** *In any nonconstant full-width transition, put $z = c_0$ and $z' = F(c)_0$. Then*

$$z' = h_9 + \mathbf{1}_{\{b=1\}} - \mathbf{1}_{\{p=9\}}, \qquad h_9 = \min(c_0, c_9).$$

If a periodic orbit has at least two zero digits in any state, its zero count is constant and at least two, and every phase is regular.

Proof. The first identity is (5) at digit zero. A mirrored difference 9 occurs precisely when a zero is paired with a nine, proving the second identity. If $p < 9$, then $h_9 = 0$ and $z' \le 1$. If $p = 9$, then

$$z' = \min(z, c_9) + \mathbf{1}_{\{b=1\}} - 1 \le z.$$

Thus the set of states with zero count at most one is forward invariant; above one the count cannot increase. Periodicity therefore forces any zero count at least two to be constant throughout the orbit. Each edge then has $p = 9$, and equality in the displayed bound forces $b = 1$. □

Lemma 8 (Regular-cycle classification)**.** *Suppose every phase of a histogram cycle is regular. Then every histogram in the cycle satisfies $\mathcal{X}$ of (10). Conversely, those inequalities describe regular cycles of period dividing three. The exact period is one precisely when $c_1 = c_2 = c_4$, and otherwise is three.*

Proof. At a regular edge the two corrections in (5) cancel, so its output has the form

$$c = (E, C_1, C_2, C_3, C_4, C_4, C_3, C_2, C_1, E + \tau), \tag{19}$$

where $E \ge 1$, $C_j \ge 0$, and $\tau \ge 0$. Here τ is the *preceding* edge's central width; its constancy has not yet been assumed. Put $H = E + C_1 + C_2 + C_3 + C_4$, so the length is $2H + \tau$. The histogram consists of a symmetric base with $2H$ digits, together with τ extra nines. Since $h_9 = \min(E, E + \tau) = E$, the quantity E remains constant at regular transitions.

Central width cannot increase. Let t be the central width computed from (19). If there is no equal central run, then $t = 0 \le \tau$. Otherwise the central run's digit j is at least 5: the symmetric base has exactly H digits below 5, and the added digits are nines. For $5 \le j \le 8$, let B be the base multiplicity of digit j and let S be the sum of the base multiplicities of digits $5, \dots, j-1$. There are $H + S$ digits below j and $H - S - B + \tau$ digits above j. Removing all unequal mirrored pairs therefore leaves

$$\begin{aligned} t &= 2H + \tau - 2\max(H + S, H - S - B + \tau) \\ &= \min(\tau - 2S,\ 2S + 2B - \tau) \le \tau. \end{aligned} \tag{20}$$

This expression is used for the actual central run, so it is nonnegative. If the median digit is 9, the added nines belong to that run rather than lying above it. There are $2H - E$ digits below 9 and none above, giving instead

$$t = 2H + \tau - 2(2H - E) = \tau - 2(H - E) \le \tau. \tag{21}$$

At the next histogram, the surplus of nines over zeros equals t. Thus, around a regular cycle, the successive surpluses form a cyclic nonincreasing sequence and are all equal. Write their common value as t from now on. Equality in (21) would imply $H = E$, hence $C_1 = C_2 = C_3 = C_4 = 0$. All positive differences would then be 9, contrary to $b = 1$. When $t > 0$, equality in (20) forces $S = 0$ and $B \ge t$. If the median digit were $j \ge 6$, symmetry and $S = 0$ would place the nearest unequal lower digit at most at $9 - j$. The smallest positive difference would be at least $2j - 9 \ge 3$. Therefore the median digit is 5. Its difference-1 multiplicity is $h_1 = C_4 - t$, so $b = 1$ implies $C_4 > t$. For $t = 0$, the histogram is symmetric, and a positive difference 1 requires the central digits 4 and 5 to occur; again $C_4 > t$.

Block overlaps force the other inequalities. Let $f(r)$ be the digit in position r of the ascending symmetric base, for $1 \le r \le H$, and set $f(r) = 0$ for $r \le 0$. The H positive mirrored differences of (19) are

$$9 - f(r) - f(r - t), \qquad 1 \le r \le H. \tag{22}$$

Indeed, the extra t nines shift the descending base string by t positions, while its symmetric base digit in position $r - t$ is $9 - f(r - t)$. The remaining central block has width t. Counting the overlaps of the constant blocks of f gives

$$h_1 = C_4 - t, \qquad h_8 = \min(C_1, t), \qquad C_1' = h_1 + h_8 = C_4 - (t - C_1)_+,$$

where $u_+ = \max(u, 0)$ and primes denote the next histogram's coordinates. If $C_1 < t$, then $C_1' > C_1$ because $C_4 > t$. If $C_1 \ge t$, then $C_1' = C_4 > t$. Consequently $\min(C_1, t)$ is nondecreasing, and increases strictly whenever $C_1 < t$. Periodicity gives $C_1 \ge t$ at every phase; applying the transition once more gives $C_1 > t$ at every phase.

Suppose now that $C_2 \le t$. With $C_1, C_4 > t$, the overlaps contributing to $h_4 + h_5$ in (22) have lower-base digit pairs $(2,2), (3,1), (3,2), (4,1)$. The potential pair $(4,0)$ is excluded by the intervening block $C_1 > t$. The four contributions sum to

$$\begin{aligned} C_4' = h_4 + h_5 = {} & (C_2 - t)_+ + \min\big(C_3, (t - C_2)_+\big) \\ & + \max\big(0, \min(C_3, t) - (t - C_2)_+\big) + (t - C_2 - C_3)_+. \end{aligned} \tag{23}$$

For example, the $(3,1)$ contribution is the overlap remaining in the digit-3 block after the intervening digit-2 block has used C_2 of the shift t; the other terms follow from the same shifted-block intersection. Put $u = t - C_2 \ge 0$. The first term vanishes, and the second and fourth terms sum to u. Thus

$$C_4' = \max\{u, \min(C_3, t)\} \le t,$$

contrary to $C_4' > t$. It follows that $C_2 > t$ throughout the cycle.

Now $C_1, C_2, C_4 > t$. Formula (22) gives the complete list

j	9	8	7	6	5	4	3	2	1
h_j	E	t	$C_1 - t$	t	$C_2 - t$	$\min(C_3, t)$	$\lvert C_3 - t\rvert$	$\min(C_3, t)$	$C_4 - t.$

(24)

Here h_3 is the sum of the within-digit-3 contribution $(C_3 - t)_+$ and the crossing digit-4/digit-2 contribution $(t - C_3)_+$; their sum is $|C_3 - t|$. Writing $\delta = (t - C_3)_+$, we obtain

$$(C_1', C_2', C_3', C_4') = (C_4, C_1 - \delta, C_3 + 2\delta, C_2 - \delta).$$

The coordinate C_3 cannot decrease, so periodicity forces $\delta = 0$ at every phase. We have proved

$$E \geq 1, \qquad C_1, C_2, C_4 \geq t + 1, \qquad C_3 \geq t,$$

which are precisely $\mathcal{X}$. Conversely, these inequalities make the same overlap computation valid with $\delta = 0$. The transition is

$$(E, t, C_1, C_2, C_3, C_4) \longmapsto (E, t, C_4, C_1, C_3, C_2).$$

It is regular and has period dividing three, with the asserted fixed-point condition. In the original word parameters of Kauers–Koutschan, the correspondence is

$$m = t, \quad e = E, \quad a = C_1 - t - 1, \quad b = C_3 - t, \quad c = C_4 - t, \quad d = C_2 - t.$$

The letter b in this last display is the published family's parameter, not the minimum mirrored difference used elsewhere. □

Lemma 9 (Uniform inner-digit imbalance)**.** *Every subtraction image satisfies*

$$|F(c)_j - F(c)_{9-j}| \leq 3, \qquad 1 \leq j \leq 4.$$

Proof. The base terms $h_j + h_{9-j}$ in (5) are symmetric, and the central term affects only digit 9. Thus all inner imbalance comes from the transfers $b \mapsto b - 1$ and $9 - p \mapsto 10 - p$. For $j = 1, 2, 3$, each adjacent-digit transfer changes the imbalance between j and $9 - j$ by at most one, giving the stronger bound two. For the pair $4, 5$, a transfer can change the imbalance by two only when it moves between those digits. The first transfer then moves $5 \mapsto 4$, while the second moves $4 \mapsto 5$; if both occur their contributions cancel. Otherwise one contribution has magnitude at most two and the other at most one. This proves the bound three. □

Lemma 10 (Coverage of the reduced query)**.** *Let Φ be the global counterexample formula defined in Section 4. Define Ψ by adding exactly*

$$\begin{gathered} c_0^{(i)} \leq 1 \quad (0 \leq i \leq 2), \\ p_0 \neq 9 \text{ or } b_0 \neq 1, \\ |c_j^{(i)} - c_{9-j}^{(i)}| \leq 3 \quad (0 \leq i \leq 2,\ 1 \leq j \leq 4). \end{gathered} \tag{25}$$

Then Φ is satisfiable if and only if Ψ is satisfiable.

Proof. Every solution of Ψ is a solution of Φ because the former adds conjuncts. Conversely, suppose Φ has a solution. By Lemma 5, it represents a three-cycle outside X and Y. If any zero count were at least two, Lemma 7 would make every phase regular, and Lemma 8 would put the cycle in X. Hence all three zero counts are at most one. The cycle must also have a nonregular phase, by the same classification lemma. Rotate that phase to index zero. The original constraints remain true after rotation: X is invariant along an orbit, and the exclusion of Y is imposed at all three phases. Finally, each histogram is a subtraction image, so Lemma 9 supplies the remaining inequalities. The rotated solution therefore satisfies Ψ. □

The reduced formula retains unbounded nonnegative digit counts; in particular, (25) imposes no bound on the total length. An unsatisfiability certificate for Ψ, combined with Lemma 10, proves the same global exclusion as an unsatisfiability certificate for Φ itself.

C Reproducibility and scope of the checks

The supplement contains the source of the arithmetic encoding, the global and reduced SMT-LIB queries, a compressed Z3 proof graph, verification reports, and independent model and counting verifiers. The canonical reduced query is `certificates/classification_builder_reduced.smt2`, whose SHA-256 is

```
dcab29251e00d29a102e95759e8c30f52
fdcf3b2a4bcdf745e32ac8b3f379eea
```

where the two displayed lines are concatenated. The checker binds its result to the query and compressed proof graph by their SHA-256 hashes. The query validator separately checks that each assertion in the reduced query's initial part follows from its corresponding global-query assertion, and that the only additions are the 28 scalar constraints expressing (25). Those checks use exact term comparison or strictly checked local implications.

C.1 What the proof graph checker verifies

A node records a Boolean conclusion and references earlier proof nodes. The verifier checks that the reachable graph is acyclic. Each node is interpreted as a sequent $\Gamma \vdash C$, where Γ is its set of open hypotheses and the exported input assertions are available globally. An asserted node must match an actual input assertion; a hypothesis node contributes its conclusion to Γ. Ordinary inference nodes inherit the union of their premises' hypothesis sets. Logical expressions are stored as shared, typed operator nodes and reconstructed by the checker; this representation adds no assumptions. Weighted Boolean constraints become exact integer sums of 0/1 indicators. Their conclusions must either pass a direct propositional or equality check or satisfy the local implication

$$(P_1 \wedge \cdots \wedge P_r) \implies C.$$

For the latter case, a fresh local check asserts the P_i and $\neg C$, without the original query assertions. It must be unsatisfiable with a complete cvc5 proof under the strict options below. Thus no inference is accepted merely because the full input was reported unsatisfiable. The direct equality checks use reflexivity, symmetry, transitivity, and congruence under identical operators with equal arguments; they do not assume an arithmetic identity without checking it.

A lemma node may discharge a hypothesis only when its premise proves false and its conclusion contains that hypothesis's negation as a clause literal. All unrecognized hypotheses remain open. The root is accepted only if its conclusion is false and its open-hypothesis set is empty. Fresh auxiliary definitions are eliminated by acyclic substitution; their introduction is not accepted as an additional input axiom. If a parsed assertion differs from the graph's asserted formula, its implication must be checked explicitly before use.

The local arithmetic checks use these explicit settings:

```
produce-proofs          true
check-proofs            true
check-proofs-complete   true
proof-allow-trust       false
proof-mode              full-proof-strict
proof-granularity       dsl-rewrite-strict
```

Every returned local proof graph is also traversed to reject trust or hole rules. The claim is a separately checked Z3 refutation using strict cvc5 local proofs. The scoped graph combines those local implications into the full contradiction. The mathematical bridge from the query to Kaprekar cycles is proved in the main text and Appendix B.

C.2 Reproduction commands

From the supplement root, install the pinned Python requirements and run the complete verification procedure:

```
python -m pip install -r requirements.txt
python src/verify_all.py
```

This runs the proof checker's rejection tests, compares the arithmetic model with direct subtraction on every histogram of lengths 1 through 8 and 2,000 larger deterministic examples, checks the family transitions and exact generating-function identity, and validates the global-to-reduced query bridge. It then checks the complete saved proof graph, including explicit assertion bridges whenever internal representations differ. The bounded examples support the encoding; its validity for all lengths is proved in Lemmas 2–5.

To regenerate the proof before verifying it, run:

```
python src/symbolic_z3proof.py --reduced --arith-solver 2
python src/verify_all.py
```

The proof-generation command may be omitted when verifying the saved graph. A timeout, `unknown` result, unverified inference or open root hypothesis is a failed verification, never a proof. The reports record solver versions, effective proof options, formula and proof hashes, and the numbers of checked nodes. All arithmetic is exact. No Lean formalization or external peer review is claimed.

Authorship. This paper and its verification programs were prepared by the AI system GPT-6 Astra. Independent AI review checked the encoding and derivation.

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